Dept. of Computer Science and Engineering (Data Science)

Adichunchanagiri Institute of Technology, Chikkamagaluru

Major Project Synopsis

TITLE: Multiple Disease Prediction

Problem Statement: Early detection of diseases like diabetes, heart disease, Parkinson's, kidney disease, and breast cancer is often delayed due to limited access to timely diagnosis. There is a need for a simple, accessible tool that can provide quick health predictions. The project develops a web-based health assistant using Streamlit and machine learning models to predict these diseases from user provided health parameters, enabling proactive health management and awareness.

Description: The multiple disease prediction system is a web-based health assistant built with Streamlit and machine learning. It predicts five major diseases-diabetes, heart disease, Parkinson's, kidney disease, and breast cancer using pre-trained models. Users can input health parameters through a simple interface and receive quick predictions. While not a substitute for medical consultation, the system provides preliminary insights that encourage timely healthcare decisions. The project highlights the role of machine learning in improving accessibility, early detection and proactive health management.

Expected Outcomes: The project is expected to deliver a reliable web application that provides quick and accurate predictions for multiple disease. Users will benefit from an interactive interface that simplifies data entry and improves accessibility. Ultimately, the system aims to support early diagnosis and promote proactive health management.

Technologies and Tools:

- **Programming Language**: Python 3.8+
- **Framework**: Streamlit (for web application)
- **Libraries**: Scikit-learn, NumPy, Pandas, TensorFlow, Pickle
- **IDE/Environment**: Jupyter Notebook / Anaconda
- **Hosting** : AWS EC2, Google Cloud, or Azure
- **Client-side**: Minimum Intel i3 / AMD Ryzen 3, 4GB RAM (8GB recommended), stable internet
- **Server-side (if deployed)**: Intel Xeon / Ryzen 7+, 16GB–32GB RAM, SSD storage, NVIDIA GPU for deep learning acceleration

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Signature of the Guide with date **Signature of the Coordinator with date**